CHARACTER SUMS WITH $\log\log q$ OUTSIDE A SPARSE SET OF CHARACTERS

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Abstract. For a primitive character χ modulo q let $M(\chi)$ be the largest partial sum of χ in modulus. We show that $M(\chi) \le c\varepsilon^{-1}\sqrt{q}\log\log q$ for all but $O\big((\log q)^C q^{\varepsilon}\big)$ primitive characters modulo q, uniformly for $0 < \varepsilon \le 1$, with absolute constants. Consequently at most $q^{O(1/\tau)}(\log q)^C$ characters have $M(\chi) > \tau\sqrt{q}\log\log q$, and all but $\exp(O((\log\log q)^2))$ characters satisfy $M(\chi) \ll \sqrt{q}\log q/\log\log q$. The bound $|L'/L(1,\chi)| \ll \varepsilon^{-1}(\log\log q + \log(2/\varepsilon))$ holds outside a set of comparable size, and a zero-free half-plane $\operatorname{Re} s > \theta$ would give both bounds for every character with a constant of order $(1-\theta)^{-1}$.

1. Introduction

Let χ be a primitive Dirichlet character modulo $q \ge 3$ and

$$M(\chi) = \max_{x \ge 1} \Big| \sum_{n \le x} \chi(n) \Big|.$$

Pólya and Vinogradov showed in 1918 that $M(\chi) \ll \sqrt{q}\log q$. Paley [13] found infinitely many quadratic characters with $M(\chi) \gg \sqrt{q}\log\log q$, and Montgomery and Vaughan [12] proved that $M(\chi) \ll \sqrt{q}\log\log q$ for every χ if the generalised Riemann hypothesis holds. Without a hypothesis, the factor $\log q$ has been lowered only for special families: Granville and Soundararajan [8] saved a power of $\log q$ for characters of fixed odd order, and Goldmakher [7] and Lamzouri and Mangerel [10] refined this. Bober, Goldmakher, Granville and Koukoulopoulos [5] determined, for prime q, the proportion of characters with $M(\chi) > \tau\sqrt{q}$ when $\tau \le \log\log q - C$; that proportion decays doubly exponentially in τ.

This paper is about the characters on the far side of that range, those with $M(\chi)$ at least a large multiple of $\sqrt{q}\log\log q$. We show that there are very few of them, with a bound that is uniform in the multiple.

Theorem 1.1. *There are absolute constants $c_1, c_2, c_3 > 0$ such that for every $q \ge 3$ and every $0 < \varepsilon \le 1$,*

$$\#\Big\{\chi \bmod q \text{ primitive} : \ M(\chi) > \frac{c_1}{\varepsilon}\sqrt{q}\log\log q\Big\} \le c_2(\log q)^{c_3} q^{\varepsilon}.$$

For a prime q there are $q-2$ primitive characters, so the bound $\sqrt{q}\log\log q$ of Montgomery and Vaughan holds, with the constant c_1/ε, for all but a proportion $q^{\varepsilon-1+o(1)}$ of them. Read the other way, Theorem 1.1 is a tail bound.

Corollary 1.2. *With the constants of Theorem 1.1, for every $q \ge 3$ and every $\tau \ge c_1$,*

$$\#\{\chi \bmod q \text{ primitive} : \ M(\chi) > \tau\sqrt{q}\log\log q\} \le c_2(\log q)^{c_3} q^{c_1/\tau}.$$

In particular all but $\exp\big(O((\log\log q)^2)\big)$ primitive characters modulo q satisfy $M(\chi) \ll \sqrt{q}\log q/\log\log q$, and all but $q^{O(1/\log\log q)}$ satisfy $M(\chi) \ll \sqrt{q}(\log\log q)^2$.

The corollary follows by taking $\varepsilon = c_1/\tau$, and then $\tau = \log q/(\log\log q)^2$ and $\tau = c_1\log\log q$. For comparison, the frequency found in [5] is $\exp(-\exp(\tau' + O(1)))$ for $M(\chi) > \tau'\sqrt{q}$ with

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$\tau' \le \log\log q - C$; at the end of that range this is q^{-c} for a constant c, and Corollary 1.2 continues it, in the units $\sqrt{q}\log\log q$, to every larger threshold.[1]

The proof of Theorem 1.1 rests on a transfer principle: the argument of Montgomery and Vaughan needs the Riemann hypothesis only through a power saving in prime sums of the twists $\chi\psi$, and a zero-free rectangle of any fixed width supplies such a saving. The point of the present version is the dependence on the width.

Theorem 1.3 (transfer)**.** *There is an absolute constant c_4 with the following property. Let χ be a primitive character modulo $q \ge 3$, let $1/2 \le \theta < 1$, put $R = \lceil \log q \rceil^{16}$, and suppose that for every $r \le R$ and every character ψ modulo r*

$$L(s, \chi\psi) \ne 0 \qquad \text{for } \operatorname{Re} s > \theta,\ |\operatorname{Im} s| \le q. \tag{1.1}$$

Then

$$M(\chi) \le \frac{c_4}{1-\theta}\sqrt{q}\log\log q.$$

The factor $(1-\theta)^{-1}$ is what makes Theorem 1.1 uniform in ε. It comes from the smooth numbers in the circle method: the major arcs need a cutoff Y with $Y^{1-\theta}$ larger than a power of $\log q$, and the harmonic sum over integers free of prime factors above Y is of size $\log Y \asymp (1-\theta)^{-1}\log\log q$. We do not know whether a smaller power of $(1-\theta)^{-1}$ is possible. The hypothesis (1.1) with $\theta = 1/2$ is part of the generalised Riemann hypothesis, so Theorem 1.3 contains the theorem of Montgomery and Vaughan. It also shows what a zero-free half-plane would give.

Corollary 1.4. *Let $1/2 \le \theta < 1$ and suppose that no Dirichlet L-function of a nonprincipal character vanishes in the half-plane $\operatorname{Re} s > \theta$. Then every primitive character χ modulo $q \ge 3$ satisfies $M(\chi) \le c_4(1-\theta)^{-1}\sqrt{q}\log\log q$ and*

$$\Big|\frac{L'}{L}(1,\chi)\Big| \le \frac{c_5}{1-\theta}\Big(\log\log q + \log\frac{2}{1-\theta}\Big),$$

with c_4 as in Theorem 1.3 and c_5 as in Theorem 1.5 below.

No such half-plane is known for any $\theta < 1$. The second bound of Corollary 1.4 is the case of every character of the following analogue of Theorem 1.1 for the logarithmic derivative at $s = 1$; under the generalised Riemann hypothesis $|L'/L(1,\chi)| \le (2+o(1))\log\log q$ for every nonprincipal χ [9].

Theorem 1.5. *There are absolute constants $c_5, c_6, c_7 > 0$ such that for every $q \ge 3$ and every $0 < \varepsilon \le 1$,*

$$\#\Big\{\chi \bmod q \text{ primitive}:\ \Big|\frac{L'}{L}(1,\chi)\Big| > \frac{c_5}{\varepsilon}\Big(\log\log q + \log\frac{2}{\varepsilon}\Big)\Big\} \le c_6\Big(\frac{\log q}{\varepsilon}\Big)^{c_7} q^{\varepsilon}.$$

Here the zero-free rectangle that is needed has height of order $\log q/\varepsilon$ instead of q, and the proof is a contour integral with a smoothed weight, in which the size of L'/L next to the zeros is controlled by the partial fraction formula.

Numerical evidence and open problems. Section 7 reports the values of $M(\chi)$ for all nonprincipal characters to prime moduli below 10^4, 5 733 937 characters in all. The ratio $M(\chi)/(\sqrt{q}\log\log q)$ exceeds 1 only for six characters, all to moduli below 12, and stays below 0.85 for $q > 1000$; for every one of the 1227 moduli the maximum is attained by an odd character; and the number of characters above a fixed multiple of $\sqrt{q}\log\log q$ drops off far faster than any tail bound of the shape of Corollary 1.2. Section 8 lists the conjectures around $M(\chi)$ and $L'/L(1,\chi)$ with what is known about each, including what the present results add (Table 2).

[1] The two results use different information. The distribution in [5] comes from the structure of characters with large sums, which must pretend to be characters of small conductor and opposite parity. Theorem 1.1 comes from the zeros of the L-functions of the twists $\chi\psi$ by characters ψ of small modulus, through a zero-density estimate.

Plan. Section 2 collects the inputs: the expansion of a character sum through Gauss sums, the explicit formula, a zero-density estimate and the estimate of Montgomery and Vaughan for exponential sums with multiplicative coefficients. Section 3 turns a zero-free rectangle into a bound for twisted prime sums. Section 4 proves Theorem 1.3, Section 5 counts the characters for which (1.1) fails and proves Theorem 1.1, and Section 6 proves Theorem 1.5 and Corollary 1.4. Figure 1 shows how the parts depend on each other.

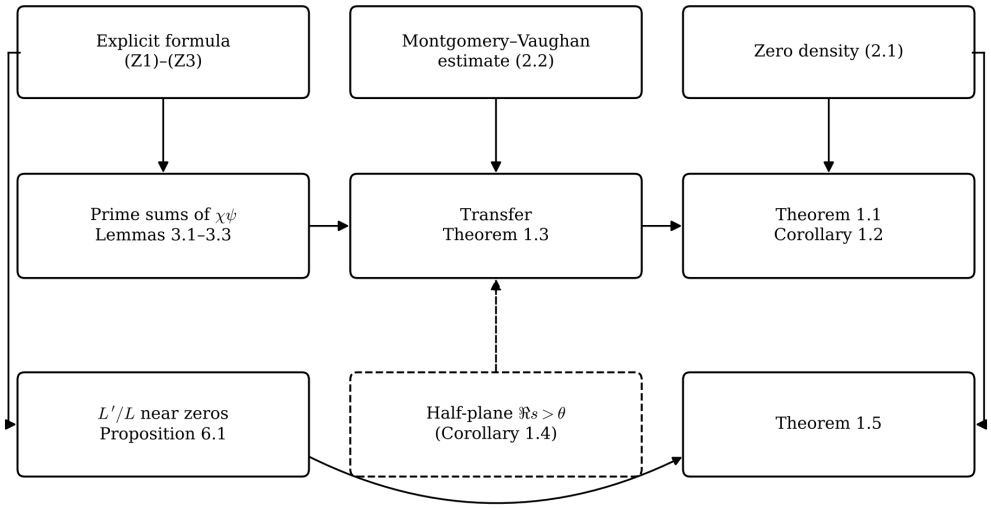


FIGURE 1. The logical structure. Solid arrows are implications proved here; the dashed arrow is the hypothetical half-plane of Corollary 1.4.

2. INPUTS

Throughout, p denotes a prime, Λ is the von Mangoldt function, $e(x) = e^{2\pi i x}$, $\|x\|$ is the distance from x to the nearest integer, and $P(n)$ is the largest prime factor of $n \geq 2$, with $P(1) = 1$. For a Dirichlet character ξ put

$$\psi(x, \xi) = \sum_{n \leq x} \xi(n)\Lambda(n), \qquad \vartheta(x, \xi) = \sum_{p \leq x} \xi(p) \log p.$$

The letters c, C, with or without subscripts, denote positive absolute constants, and $A \ll B$ means $|A| \leq CB$ with such a constant. A zero of an L-function always means a zero in the strip $0 \leq \operatorname{Re} s \leq 1$, written $\rho = \beta + i\gamma$.

2.1. **Character sums through Gauss sums.**

Lemma 2.1. *Let χ be primitive modulo $q \geq 3$, $H = \lfloor q/2 \rfloor$, and let $B \geq 0$ be such that*

$$\Big| \sum_{m \leq H} \frac{\bar{\chi}(m) e(m\beta)}{m} \Big| \leq B \qquad \text{for every } \beta \in \mathbb{R}.$$

Then $M(\chi) \leq \sqrt{q}\,(B + 2)$.

Proof. Let $\tau = \sum_{a \bmod q} \bar{\chi}(a) e(a/q)$. For primitive $\bar{\chi}$ one has $\sum_a \bar{\chi}(a) e(an/q) = \chi(n)\tau$ for every integer n and $|\tau| = \sqrt{q}$ [6, Chapter 9]. It suffices to bound $\sum_{n \leq T} \chi(n)$ for integers $T \geq 1$. Summing the first identity over $1 \leq n \leq T$ and writing the residues as $-q/2 < a \leq q/2$,

$$\tau \sum_{n \leq T} \chi(n) = \sum_{0 < |a| \leq q/2} \bar{\chi}(a) \big(e(aT/q) - 1\big)\, g(a/q), \qquad g(\vartheta) = \frac{1}{1 - e(-\vartheta)},$$

since the geometric sum is $\sum_{n=1}^{T} e(n\vartheta) = e(\vartheta)\frac{e(T\vartheta)-1}{e(\vartheta)-1} = \frac{e(T\vartheta)-1}{1-e(-\vartheta)}$, and $a=0$ contributes nothing because $\bar\chi(0)=0$. With $x=2\pi\vartheta$, $0<|x|\le\pi$, one has $g(\vartheta)=\frac12+\frac{i}{2}\cot\frac{x}{2}$, so

$$g(\vartheta)=\frac{1}{2\pi i\vartheta}+r(\vartheta),\qquad r(\vartheta)=\frac12+\frac{i}{2}\Big(\cot\frac{x}{2}-\frac{2}{x}\Big).$$

On $0<|y|\le\pi/2$ the odd function $\cot y-1/y$ has modulus at most $2/\pi$, its value at $|y|=\pi/2$, because it is decreasing on $(0,\pi/2]$ and tends to 0 at 0. Hence $|r(\vartheta)|\le(1/4+1/\pi^2)^{1/2}<0.6$, and the terms with r contribute at most $2\cdot0.6\cdot q$. In the remaining terms put $a=m$ or $a=-m$ with $1\le m\le H$ (when q is even the residue $q/2$ is not prime to q and has $\bar\chi(q/2)=0$). They equal

$$\frac{q}{2\pi i}\sum_{m\le H}\frac{\bar\chi(m)}{m}\Big(e\Big(\frac{mT}{q}\Big)-1-\bar\chi(-1)\Big(e\Big(-\frac{mT}{q}\Big)-1\Big)\Big),$$

a combination with coefficients of modulus 1 of four sums of the kind in the hypothesis, so they contribute at most $4qB/(2\pi)$. Dividing by $|\tau|=\sqrt q$ gives $|\sum_{n\le T}\chi(n)|\le\sqrt q\,(2B/\pi+1.2)$. □

Lemma 2.2. *Let χ be primitive modulo q, let $1\le r<q$ and let ψ be a character modulo r. Then $\chi\psi$, as a character modulo qr, is nonprincipal. If χ' is another primitive character modulo q with $\chi'\psi=\chi\psi$, then $\chi'=\chi$.*

Proof. Let $d=(q,r)$. Every n prime to q with $n\equiv1 \pmod d$ has, by the Chinese remainder theorem, a representative $n'\equiv n \pmod q$ with $n'\equiv1 \pmod r$, and n' is prime to qr. If $\chi\psi$ were principal, then $\chi(n)=\chi(n')=\chi(n')\psi(n')=1$ for all such n, so χ would be trivial on the kernel of $(\mathbb{Z}/q\mathbb{Z})^\times\to(\mathbb{Z}/d\mathbb{Z})^\times$ and hence induced by a character modulo $d\le r<q$, which is impossible for a primitive character. If $\chi'\psi=\chi\psi$, then χ' and χ agree at every integer prime to qr; every residue class prime to q contains such an integer, again by the Chinese remainder theorem, so $\chi'=\chi$. □

2.2. **Zeros and the explicit formula.** We use the following facts for a primitive character ξ modulo $k\ge3$; all are in [6], Chapters 16 and 19.

(Z1) For every real t the number of zeros of $L(s,\xi)$ with $|\gamma-t|\le1$ is at most $c_8\log(k(|t|+2))$.

(Z2) For $-1\le\sigma\le2$ and $s=\sigma+it$,

$$\frac{L'}{L}(s,\xi)=\sum_{|\gamma-t|<1}\frac{1}{s-\rho}+O\big(\log(k(|t|+2))\big).$$

(Z3) For $x\ge2$ and $2\le T\le x$,

$$\psi(x,\xi)=-\sum_{|\gamma|<T}\frac{x^\rho}{\rho}-(1-a)\log x-b(\xi)+\sum_{m\ge1}\frac{x^{a-2m}}{2m-a}+O\Big(\frac{x\log^2(kx)}{T}+x^{1/4}\log x\Big),$$

where $a\in\{0,1\}$ is given by $\xi(-1)=(-1)^a$ and $b(\xi)$ is a constant with

$$b(\xi)=-\sum_{|\gamma|<1}\frac{1}{\rho}+O(\log k).$$

The zero-density estimate we use is Montgomery's [11, Theorem 12.2]: for $k\ge1$, $T\ge2$ and $4/5\le\sigma\le1$,

$$\sum_{\xi \bmod k}N(\sigma,T,\xi)\le c_9\,(kT)^{2(1-\sigma)/\sigma}(\log kT)^{14},\tag{2.1}$$

where $N(\sigma,T,\xi)$ is the number of zeros of $L(s,\xi)$ with $\beta\ge\sigma$ and $|\gamma|\le T$, and the sum runs over all characters modulo k.[2]

[2]For an imprimitive ξ the zeros in $0<\operatorname{Re}s\le1$ are those of the primitive character inducing it, since the Euler factors that differ vanish only on $\operatorname{Re}s=0$. Only the shape of (2.1) matters below: the exponent of kT is at most $\frac52(1-\sigma)$, and the logarithmic factor is a fixed power. A log-free estimate of Gallagher's type would remove the factor $(\log q)^{c_3}$ in Theorem 1.1 up to the contribution of the moduli $r\le R$.

2.3. **Exponential sums with multiplicative coefficients.** The minor arcs are handled by an estimate of Montgomery and Vaughan [12, Corollary 1]: there is an absolute constant c_{10} such that, if f is multiplicative with $|f(n)| \le 1$, a, r are coprime with $|\alpha - a/r| \le r^{-2}$, and $2 \le R \le r \le N/R$, then

$$\Big|\sum_{n\le N} f(n)e(n\alpha)\Big| \le c_{10}\Big(\frac{N}{\log N} + \frac{N(\log R)^{3/2}}{R^{1/2}}\Big). \tag{2.2}$$

2.4. **Two elementary bounds.**

Lemma 2.3. *There is an absolute constant c_{11} such that for $Y \ge 2$ and every $H \ge 1$*

$$\sum_{\substack{n\le H\\ P(n)\le Y}} \frac{1}{n} \le \prod_{p\le Y}\Big(1-\frac{1}{p}\Big)^{-1} \le c_{11}\log Y.$$

Proof. Expanding each factor as a geometric series shows that the product is the sum of $1/n$ over all n with $P(n) \le Y$; the second inequality is Mertens' theorem. □

Lemma 2.4. *For $3/4 \le u \le 2$ and real t, $|\Gamma(u+it)| \le 15(1+t^2)^{-1}$. For $-1/4 \le u \le 2$ and $|t| \ge 1$, $|\Gamma(u+it)| \le c_{12}|t|^{3/2}e^{-\pi|t|/2}$.*

Proof. The product formula gives $|\Gamma(u+it)|^2 = \Gamma(u)^2\prod_{n\ge0}\big(1+t^2/(u+n)^2\big)^{-1}$. Keeping the factors $n = 1, 2$ and using $u+1 \le 3$, $u+2 \le 4$,

$$|\Gamma(u+it)| \le \Gamma(u)\Big(1+\frac{t^2}{9}\Big)^{-1/2}\Big(1+\frac{t^2}{16}\Big)^{-1/2} \le \Gamma(u)\Big(1+\frac{t^2}{12}\Big)^{-1} \le \frac{12\,\Gamma(u)}{12+t^2},$$

by the inequality of arithmetic and geometric means, and $\Gamma(u) \le \Gamma(3/4) < 1.23$ on $[3/4, 2]$; so the left side is at most $14.8/(12+t^2) \le 15/(1+t^2)$. The second is Stirling's formula, uniform for u in a compact interval and $|t| \ge 1$ [14, §4.42]. □

3. Prime sums from a zero-free rectangle

In this section $1/2 \le \theta < 1$ is fixed and ξ is a nonprincipal character modulo $k \ge 3$. The constant c_{13} below does not depend on θ; this is what keeps the final constant of order $(1-\theta)^{-1}$.

Lemma 3.1. *Let $x \ge 2$ and suppose that $L(s,\xi) \ne 0$ for $\operatorname{Re} s > \theta$, $|\operatorname{Im} s| \le x$. Then*

$$|\psi(x,\xi)| \le c_{13}\,x^{\theta}\log^2(kx) \qquad \text{and} \qquad |\vartheta(x,\xi)| \le c_{13}\,x^{\theta}\log^2(kx).$$

Proof. Let ξ^* be the primitive character modulo $k^* \mid k$ that induces ξ; then $k^* \ge 3$, since ξ is nonprincipal and there is no primitive character modulo 2. The two characters differ only at integers sharing a prime factor with k, so

$$|\psi(x,\xi) - \psi(x,\xi^*)| \le \sum_{p\mid k}\sum_{p^m\le x}\log p \le \omega(k)\log x \le \frac{\log k\log x}{\log 2},$$

and $L(s,\xi)$, $L(s,\xi^*)$ have the same zeros with $\operatorname{Re} s > 0$. Apply (Z3) to ξ^* with $T = x$. Every zero with $|\gamma| < x$ has $\beta \le \theta$ by hypothesis. The zeros with $|\gamma| < 1$ are combined with $b(\xi^*)$:

$$-\sum_{|\gamma|<1}\frac{x^{\rho}}{\rho} - b(\xi^*) = -\sum_{|\gamma|<1}\frac{x^{\rho}-1}{\rho} + O(\log k),$$

$$\Big|\frac{x^{\rho}-1}{\rho}\Big| = \Big|\int_0^{\log x} e^{\rho u}\,du\Big| \le x^{\beta}\log x \le x^{\theta}\log x,$$

and by (Z1) there are at most $c_8\log(2k)$ of them. The zeros with $j \le |\gamma| < j+1$, $1 \le j < x$, number at most $2c_8\log(k(j+3))$ and contribute at most x^{θ}/j each, so together at most $2c_8x^{\theta}\log(k(x+3))\sum_{j\le x}1/j \ll x^{\theta}\log^2(kx)$. The terms $(1-a)\log x$ and $\sum_m x^{a-2m}/(2m-a) \le \sum_m x^{1-2m} \le 1$ are smaller, and the error term of (Z3) is $O(\log^2(kx) + x^{1/4}\log x)$.

Since $\theta \ge 1/2$, all of this is $O(x^\theta \log^2(kx))$. Finally $\psi(x,\xi) - \vartheta(x,\xi)$ is a sum over prime powers $p^m \le x$ with $m \ge 2$, of modulus at most $\sum_{2 \le m \le \log x/\log 2} \vartheta(x^{1/m}, \xi_0) \ll \sqrt{x}\log x$ by Chebyshev's bound. □

Lemma 3.2. *Let $2 \le A' < B' \le x$, let $\gamma \in \mathbb{R}$, and suppose that $L(s,\xi) \ne 0$ for $\operatorname{Re} s > \theta$, $|\operatorname{Im} s| \le x$. Then*

$$\Big| \sum_{A' < p \le B'} \xi(p) e(p\gamma) \Big| \le c_{13} \, \frac{(3 + 2\pi|\gamma|B')\, B'^{\theta} \log^2(kB')}{\log A'}.$$

Proof. Write $V = \max_{2 \le u \le B'} |\vartheta(u,\xi)|$, which is at most $c_{13} B'^{\theta} \log^2(kB')$ by Lemma 3.1. By partial summation,

$$\sum_{A' < p \le B'} \xi(p) e(p\gamma) = \Big[\vartheta(u,\xi) \frac{e(u\gamma)}{\log u} \Big]_{A'}^{B'} - \int_{A'}^{B'} \vartheta(u,\xi) \frac{d}{du}\Big(\frac{e(u\gamma)}{\log u} \Big) du.$$

The derivative has modulus at most $2\pi|\gamma|/\log u + 1/(u \log^2 u)$, whose integral over $[A', B']$ is at most $(2\pi|\gamma|B' + 1)/\log A'$. The boundary terms are at most $2V/\log A'$. □

The next lemma is the major-arc estimate. Here and later χ is primitive modulo $q \ge 3$.

Lemma 3.3. *Let $1 \le r < q$, $b \in \mathbb{Z}$, $\beta \in \mathbb{R}$, let $Y \ge \max(2, r)$ and let $t \le q$ be a positive integer. Suppose that $L(s, \chi\psi) \ne 0$ for $\operatorname{Re} s > \theta$, $|\operatorname{Im} s| \le q$, for every character ψ modulo r. Then*

$$\Big| \sum_{\substack{n \le t \\ P(n) > Y}} \chi(n)\, e\Big(n\Big(\frac{b}{r} + \beta \Big) \Big) \Big| \le c_{13} \, \frac{r\,(3 + 2\pi|\beta|t) \log^2(qrt)}{(1-\theta)\log Y} \cdot \frac{t}{Y^{1-\theta}}.$$

Proof. Each $n \le t$ with $P(n) > Y$ factors uniquely as $n = pm$ with $p = P(n)$; then $p > Y$, $p \ge P(m)$ and $m \le t/p < t/Y$. Conversely, for $m < t/Y$ the admissible p are the primes in $(A_m, t/m]$ with $A_m = \max(Y, P(m) - 1)$. Since $p > Y \ge r$, p is prime to r, and on the reduced residues modulo r

$$e\Big(\frac{umb}{r}\Big) = \sum_{\psi \bmod r} c_\psi(m)\psi(u), \qquad c_\psi(m) = \frac{1}{\varphi(r)} \sum_{\substack{u \bmod r \\ (u,r)=1}} \bar\psi(u) e\Big(\frac{umb}{r}\Big), \quad |c_\psi(m)| \le 1.$$

Hence the sum to be estimated equals

$$\sum_{m < t/Y} \chi(m) \sum_{\psi \bmod r} c_\psi(m) \sum_{A_m < p \le t/m} (\chi\psi)(p)\, e(pm\beta).$$

By Lemma 2.2 each $\chi\psi$ is a nonprincipal character modulo qr, and by hypothesis it has no zero with $\operatorname{Re} s > \theta$, $|\operatorname{Im} s| \le q$. Lemma 3.2 with $k = qr$, $x = q$, $\gamma = m\beta$, $A' = A_m \ge Y$ and $B' = t/m$ bounds the inner sum by $c_{13}(3 + 2\pi|\beta|t)(t/m)^\theta \log^2(qrt)/\log Y$. There are $\varphi(r) \le r$ characters ψ, and

$$\sum_{m < t/Y} m^{-\theta} \le \int_0^{t/Y} u^{-\theta}\, du = \frac{(t/Y)^{1-\theta}}{1-\theta}.$$

Multiplying gives the bound, since $t^\theta (t/Y)^{1-\theta} = tY^{\theta-1}$. □

Remark 3.4. The factor $(1-\theta)^{-1}$ in Lemma 3.3 is harmless: it is cancelled in the next section by choosing $\log Y$ of size $(1-\theta)^{-1}\log\log q$. That choice is what produces the factor $(1-\theta)^{-1}$ of Theorem 1.3, through the smooth numbers below Y.

4. Proof of the transfer theorem

Let χ be primitive modulo $q \geq 3$ and $1/2 \leq \theta < 1$. Put

$$L = \log q, \qquad M = \lceil L \rceil, \qquad R = M^{16}, \qquad A = \Big\lceil \frac{40}{1-\theta} \Big\rceil, \qquad Y = M^A.$$

Let $L_0 \geq e^e$ be an absolute constant, large enough for the finitely many numerical inequalities used below, among them $R < q$ and $72c_{13}M^{-6}(1+L) + c_{10}(16\log M)^{3/2}M^{-8}(1+L) \leq 1$ for $L \geq L_0$.

Proposition 4.1. *Suppose that $L \geq L_0$ and that (1.1) holds for every $r \leq R$ and every character ψ modulo r. Then for every integer $1 \leq H \leq q$ and every $\beta \in \mathbb{R}$*

$$\Big| \sum_{m \leq H} \frac{\chi(m) e(m\beta)}{m} \Big| \leq \frac{c_{14}}{1-\theta} \log\log q.$$

Proof. Split the sum according to $P(m) \leq Y$ or $P(m) > Y$. The first part is at most $c_{11}\log Y$ by Lemma 2.3. For the second let

$$\mathcal{A}(t) = \sum_{\substack{n \leq t \\ P(n) > Y}} \chi(n) e(n\beta), \qquad w(t) = \frac{1}{t} - \frac{1}{t+1}.$$

Partial summation and $|\mathcal{A}(H)| \leq H$ give

$$\Big| \sum_{\substack{m \leq H \\ P(m) > Y}} \frac{\chi(m) e(m\beta)}{m} \Big| \leq 1 + \sum_{1 \leq t < H} |\mathcal{A}(t)|\, w(t). \tag{4.1}$$

Since $A \geq 80$, $Y \geq R$, and $\mathcal{A}(t) = 0$ for $t \leq Y$. Fix an integer t with $Y < t < H$ and put $Q = \lfloor t/R \rfloor \geq 1$. By Dirichlet's approximation theorem there are coprime integers a, r with $1 \leq r \leq Q$ and $|\beta - a/r| \leq 1/(r(Q+1))$. As $Q + 1 > t/R$, this gives $|\beta - a/r|\, t \leq R$.

Major arcs, $r \leq R$. Here $r \leq R < q$ and $Y \geq \max(2, r)$, and the hypothesis of Lemma 3.3 is (1.1) for this r. That lemma, with $b = a$ and $\beta - a/r$ in place of β, gives

$$|\mathcal{A}(t)| \leq c_{13}\, \frac{R\,(3 + 2\pi R)\log^2(qRt)}{(1-\theta)\log Y} \cdot \frac{t}{Y^{1-\theta}}.$$

Now $3 + 2\pi R \leq 8M^{16}$, $\log(qRt) \leq 2L + 16\log M \leq 3M$, $Y^{1-\theta} = M^{A(1-\theta)} \geq M^{40}$ and $(1-\theta)\log Y = (1-\theta)A\log M \geq 40\log M \geq 1$. Hence $|\mathcal{A}(t)| \leq 72c_{13}\, tM^{-6}$.

Minor arcs, $r > R$. Here $R < r \leq Q \leq t/R$ and $|\beta - a/r| \leq 1/(r(Q+1)) \leq r^{-2}$, so (2.2) with $f = \chi$ and $N = t$ applies to the complete sum $\sum_{n \leq t} \chi(n) e(n\beta)$. Removing the $\Psi(t, Y) = \#\{n \leq t : P(n) \leq Y\}$ terms with $P(n) \leq Y$,

$$|\mathcal{A}(t)| \leq c_{10}\Big(\frac{t}{\log t} + t\varrho\Big) + \Psi(t, Y), \qquad \varrho = \frac{(\log R)^{3/2}}{R^{1/2}} = \frac{(16\log M)^{3/2}}{M^8}.$$

Summation over t. Since $t\, w(t) \leq 1/t$, the two cases together give

$$\sum_{1 \leq t < H} |\mathcal{A}(t)|\, w(t) \leq (72c_{13}M^{-6} + c_{10}\varrho) \sum_{t \leq q} \frac{1}{t} + c_{10} \sum_{2 \leq t \leq q} \frac{1}{t\log t} + \sum_{1 \leq t < H} \Psi(t, Y)\, w(t).$$

The first term is at most 1 by the choice of L_0, since $\sum_{t \leq q} 1/t \leq 1 + L$. The second is at most $c_{10}(\log L + 2)$. In the third, each $n \leq H - 1$ with $P(n) \leq Y$ is counted by $\Psi(t, Y)$ for $n \leq t < H$, so the term equals $\sum_{n < H,\, P(n) \leq Y} (1/n - 1/H) \leq c_{11}\log Y$. With (4.1) and the smooth part,

$$\Big| \sum_{m \leq H} \frac{\chi(m) e(m\beta)}{m} \Big| \leq 2c_{11}\log Y + 2 + c_{10}(\log L + 2).$$

Finally $\log Y = A\log M \leq (\frac{40}{1-\theta} + 1)\log(2L) \leq \frac{82}{1-\theta}\log L$, because $L \geq e^e$. The proposition follows with $c_{14} = 164c_{11} + 3c_{10} + 2$, using $\log L = \log\log q \geq 1$. □

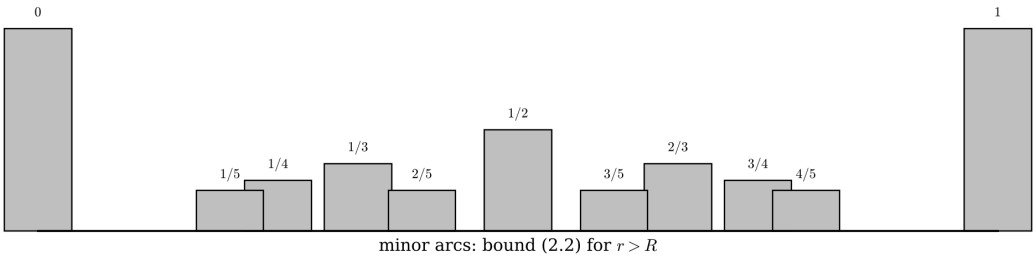


FIGURE 2. The dissection of $\mathbb{R}/\mathbb{Z}$ at a fixed t, drawn with $R = 5$. Near a fraction a/r with $r \le R$ the sum $\mathcal{A}(t)$ is a combination of prime sums of the twists $\chi\psi$, $\psi \bmod r$, and is small because of the zero-free rectangles (1.1); away from such fractions it is small by (2.2). The arcs have width $2R/t$.

Proof of Theorem 1.3. If $L < L_0$, then every sum $\sum_{m\le H} \bar\chi(m)e(m\beta)/m$ with $H \le q$ has modulus at most $1+\log q < 1+L_0$, and Lemma 2.1 gives $M(\chi) \le (3+L_0)\sqrt{q} \le \frac{3+L_0}{\log\log 3}\sqrt{q}\log\log q$ without any hypothesis. Let $L \ge L_0$. The conjugate character $\bar\chi$ is primitive modulo q and satisfies (1.1) for every $r \le R$ and $\psi \bmod r$, because the zeros of $L(s,\bar\chi\psi)$ are the complex conjugates of those of $L(s,\chi\bar\psi)$ and the rectangle is symmetric about the real axis. Proposition 4.1 for $\bar\chi$ with $H = \lfloor q/2\rfloor$ and Lemma 2.1 give

$$M(\chi) \le \sqrt{q}\Big(\frac{c_{14}}{1-\theta}\log\log q + 2\Big) \le \frac{c_{14}+2}{1-\theta}\sqrt{q}\log\log q.$$

Theorem 1.3 holds with $c_4 = \max\big(c_{14}+2, (3+L_0)/\log\log 3\big)$. □

5. Counting the exceptional characters

Proof of Theorem 1.1. Let $0 < \varepsilon \le 1$, put $\theta = 1-\varepsilon/8$, so that $7/8 \le \theta < 1$, and let $c_1 = 8c_4$ with c_4 the constant of Theorem 1.3. Let $L = \log q$, $M = \lceil L\rceil$ and $R = M^{16}$. If $L < L_0$, the proof of Theorem 1.3 gives $M(\chi) \le c_4\sqrt{q}\log\log q$ for every χ without any hypothesis, and there is nothing to count. Let $L \ge L_0$ and let E_q be the set of primitive characters χ modulo q for which (1.1) fails for some $r \le R$ and some character ψ modulo r. Every primitive $\chi \notin E_q$ satisfies

$$M(\chi) \le \frac{c_4}{1-\theta}\sqrt{q}\log\log q = \frac{c_1}{\varepsilon}\sqrt{q}\log\log q$$

by Theorem 1.3, so it suffices to bound $|E_q|$.

For $\chi \in E_q$ choose $r \le R$, $\psi \bmod r$ and a zero ρ of $L(s,\chi\psi)$ with $\beta > \theta$ and $|\gamma| \le q$; it is counted by $N(\theta,q,\chi\psi)$. For a fixed pair (r,ψ) the map $\chi \mapsto \chi\psi$ into the characters modulo qr is injective by Lemma 2.2, since $r \le R < q$. Hence, by (2.1) with $k = qr$ and $T = q$,

$$|E_q| \le \sum_{r\le R}\sum_{\psi \bmod r}\sum_{\xi \bmod qr} N(\theta,q,\xi) \le c_9\sum_{r\le R}\varphi(r)\,(q^2r)^{2(1-\theta)/\theta}\big(\log q^2r\big)^{14}.$$

Here $2(1-\theta)/\theta \le \frac{16}{7}(1-\theta) = \frac{2}{7}\varepsilon$, so $(q^2r)^{2(1-\theta)/\theta} \le q^{4\varepsilon/7}R^{2/7}$. With $R \le (2L)^{16}$, $R^{2/7} \le (2L)^5$, $\sum_{r\le R}\varphi(r) \le R^2$ and $\log q^2R \le 3L$,

$$|E_q| \le c_9\,3^{14}\,(2L)^{37}L^{14}\,q^{4\varepsilon/7} \le c_2(\log q)^{51}q^{\varepsilon},$$

with $c_2 = 2^{37}3^{14}c_9$. This proves Theorem 1.1 with $c_3 = 51$. □

Proof of Corollary 1.2. For $\tau \ge c_1$ apply Theorem 1.1 with $\varepsilon = c_1/\tau \le 1$. For the two particular cases take $\tau = \log q/(\log\log q)^2$, where $q^{c_1/\tau} = \exp(c_1(\log\log q)^2)$, and $\tau = c_1\log\log q$, where $q^{c_1/\tau} = q^{1/\log\log q}$; in both cases $(\log q)^{c_3}$ is of smaller order, and for the finitely many q with $\tau < c_1$ the statements are trivial. □

Remark 5.1. The exponent ε enters only through the width $1-\theta$ of the zero-free rectangle, and the factor $(\log q)^{51}$ only through the number of twists and the logarithmic factor in (2.1). A zero-density estimate with exponent $c(1-\sigma)$ for any absolute c gives Theorem 1.1 with $\theta = 1-\varepsilon/(4c)$, and the moduli $r \le R$ can be taken smaller, $R = (\log q)^4$ for instance, at the cost of a larger L_0, which lowers the power of $\log q$.

6. The logarithmic derivative at $s = 1$

Proposition 6.1. *Let χ be primitive modulo $q \ge 3$, let $1/2 \le \theta < 1$ and $\delta = (1-\theta)/2$, put $L = \log q$ and*

$$T_* = \frac{\max(1, \log L)}{\delta} + L + \log\frac{1}{\delta} + T_0 + 1,$$

where T_0 is an absolute constant fixed in the proof. If $L(s,\chi) \ne 0$ for $\operatorname{Re} s > \theta$, $|\operatorname{Im} s| \le T_*$, then

$$\Big|\frac{L'}{L}(1,\chi)\Big| \le \frac{c_{16}}{1-\theta}\Big(\log\log q + \log\frac{2}{1-\theta}\Big).$$

Proof. Let x be given by $\log x = \max(1, \log L)/\delta$, so that $x \ge e^4$ and $x^{-\delta}L \le 1$, and let $T = T_* - 1$. For $c > 0$ and $y > 0$ one has $e^{-y} = \frac{1}{2\pi i}\int_{(c)} \Gamma(s) y^{-s}\, ds$, and since $\sum_n \chi(n)\Lambda(n) n^{-1-s} = -\frac{L'}{L}(1+s,\chi)$ converges absolutely for $\operatorname{Re} s = 1$,

$$\Sigma := \sum_{n\ge 1} \frac{\chi(n)\Lambda(n)}{n} e^{-n/x} = \frac{1}{2\pi i}\int_{(1)} \Gamma(s) x^s \Big(-\frac{L'}{L}(1+s,\chi)\Big) ds. \tag{6.1}$$

Replace the part $|\operatorname{Im} s| \le T$ of the line $\operatorname{Re} s = 1$ by the path from $1 - iT$ to $-\delta - iT$, then to $-\delta + iT$, then to $1 + iT$ (Figure 3). Between the two paths the integrand has a pole at $s = 0$, from Γ, with residue $-\frac{L'}{L}(1,\chi)$, and no other pole: Γ has no other pole with $\operatorname{Re} s > -1$, $L(s,\chi)$ is entire, and $L(1+s,\chi)$ has no zero with $\operatorname{Re}(1+s) \ge 1-\delta = \theta + \delta$ and $|\operatorname{Im} s| \le T$. Hence

$$\frac{L'}{L}(1,\chi) = -\Sigma + J_{\text{left}} + J_{\text{hor}} + J_{\text{far}},$$

the three integrals of $\frac{1}{2\pi i}\Gamma(s)x^s(-\frac{L'}{L}(1+s,\chi))$ over the vertical segment $\operatorname{Re} s = -\delta$, $|\operatorname{Im} s| \le T$, over the two horizontal segments, and over $\operatorname{Re} s = 1$, $|\operatorname{Im} s| \ge T$.

The size of L'/L. Let $1 + s = \sigma + it$ with $\theta + \delta \le \sigma \le 2$ and $|t| \le T$. The zeros in (Z2) have $|\gamma| < T + 1 = T_*$, hence $\beta \le \theta$ and $|1 + s - \rho| \ge \sigma - \beta \ge \delta$; by (Z1) there are at most $c_8 \log(q(|t|+2))$ of them, so

$$\Big|\frac{L'}{L}(1+s,\chi)\Big| \le \frac{C\log(q(|t|+2))}{\delta}. \tag{6.2}$$

The left side. Here $s = -\delta + it$, $|x^s| = x^{-\delta}$, and by Lemma 2.4, since $3/4 \le 1 - \delta < 1$,

$$|\Gamma(-\delta + it)| = \frac{|\Gamma(1-\delta+it)|}{|-\delta+it|} \le \frac{15}{(1+t^2)\max(\delta, |t|)}.$$

With (6.2) and $\log(q(|t|+2)) \le L + \log(|t|+2)$,

$$|J_{\text{left}}| \le \frac{Cx^{-\delta}}{\delta}\int_{\mathbb{R}} \frac{L + \log(|t|+2)}{(1+t^2)\max(\delta,|t|)}\, dt \le \frac{Cx^{-\delta}L}{\delta}\Big(1 + \log\frac{1}{\delta}\Big) \le \frac{C}{\delta}\Big(1 + \log\frac{1}{\delta}\Big),$$

because the integral over $|t| \le \delta$ is at most $2\log(3q) \le 4L$ and the integral over $|t| > \delta$ is at most $2\int_\delta^\infty (L + \log(t+2))\,\frac{dt}{t(1+t^2)} \ll L(1 + \log(1/\delta))$.

The horizontal sides and the tail. On the horizontal sides $s = u \pm iT$ with $-\delta \le u \le 1$; there $|x^s| \le x$, $|\Gamma(s)| \le c_{12} T^{3/2} e^{-\pi T/2}$ by Lemma 2.4 (as $T \ge 1$ and $u \ge -1/4$), and (6.2) applies. On the tail $|L'/L(2+it,\chi)| \le \sum_n \Lambda(n) n^{-2} < 1$ and $|\Gamma(1+it)| \le c_{12}|t|^{3/2} e^{-\pi|t|/2}$. Hence

$$|J_{\text{hor}}| + |J_{\text{far}}| \le C\,\frac{x\,T^{5/2}\log(q(T+2))}{\delta}\, e^{-\pi T/2} \le C\,T^{5/2}(L+T)\,e^{-L-T/2} \le 1,$$

where we used $\pi T/2 \geq T + T/2 \geq \log x + L + \log(1/\delta) + T/2$, and $T \geq T_0$ with T_0 chosen so that $C\,T^{5/2}(1+T)e^{-T/2} \leq 1$ for $T \geq T_0$ (note $Le^{-L} \leq 1$).

Conclusion. Since $|\chi(n)| \leq 1$, $|\Sigma| \leq \sum_{n\leq x} \Lambda(n)/n + \sum_{k\geq 1} e^{-k} \sum_{kx<n\leq(k+1)x} \Lambda(n)/n \leq \log x + c_{15}$ by Mertens' estimate $\sum_{n\leq x} \Lambda(n)/n = \log x + O(1)$ and Chebyshev's bound $\psi(y) \ll y$. Collecting,

$$\Big|\frac{L'}{L}(1,\chi)\Big| \leq \frac{\max(1,\log L)}{\delta} + c_{15} + \frac{C}{\delta}\Big(1 + \log\frac{1}{\delta}\Big) + 1,$$

and $1/\delta = 2/(1-\theta)$, $\log(1/\delta) = \log(2/(1-\theta)) \geq \log 4$, $\max(1,\log L) \leq \log\log q + \log(2/(1-\theta))$ give the claim. □

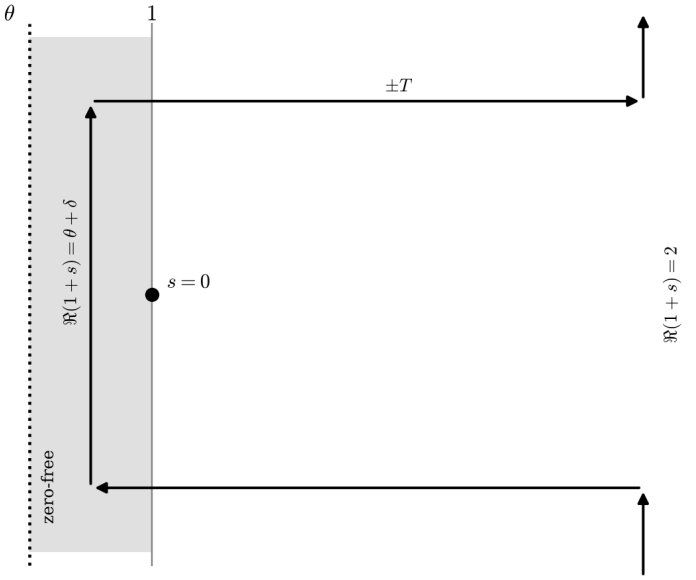


FIGURE 3. The contour in the proof of Proposition 6.1, in the variable $1+s$. The shaded region is free of zeros of $L(s,\chi)$ by hypothesis; the path keeps the distance δ from it. Beyond height T the path stays on $\mathrm{Re}(1+s) = 2$, where the factor $\Gamma(s)$ is exponentially small.

Proof of Theorem 1.5. Let $0 < \varepsilon \leq 1$, $\theta = 1 - \varepsilon/5 \geq 4/5$ and $\delta = \varepsilon/10$. A primitive χ for which $|L'/L(1,\chi)|$ exceeds the bound of Proposition 6.1 has a zero with $\beta > \theta$ and $|\gamma| \leq T_*$. Now $T_* \leq C\log q/\varepsilon$, so by (2.1) with $k = q$ and $T = T_*$ the number of such χ is at most

$$c_9\,(qT_*)^{2(1-\theta)/\theta}(\log qT_*)^{14} \leq c_9\,q^{\varepsilon/2}\Big(\frac{C\log q}{\varepsilon}\Big)^{1/2}\Big(\log\frac{Cq\log q}{\varepsilon}\Big)^{14} \leq c_6\Big(\frac{\log q}{\varepsilon}\Big)^{c_7} q^{\varepsilon}.$$

For the other characters Proposition 6.1 gives $|L'/L(1,\chi)| \leq \frac{5c_{16}}{\varepsilon}(\log\log q + \log\frac{10}{\varepsilon}) \leq \frac{c_5}{\varepsilon}(\log\log q + \log\frac{2}{\varepsilon})$ with $c_5 = 20c_{16}$, since $\log(10/\varepsilon) \leq 4\log(2/\varepsilon)$. □

Proof of Corollary 1.4. Let χ be primitive modulo q. If $\log q < L_0$, the bound for $M(\chi)$ holds unconditionally, as in the proof of Theorem 1.3. Otherwise $R < q$, every $\chi\psi$ with $r \leq R$ is nonprincipal by Lemma 2.2, and the hypothesis gives (1.1); Theorem 1.3 applies. The character χ itself is nonprincipal, so Proposition 6.1 applies as well, and $c_{16} \leq c_5$. □

7. Numerical data

For every prime $5 \le q < 10^4$ and every nonprincipal character χ modulo q we computed $M(\chi)$ exactly in the following sense. With g the least primitive root modulo q and $\operatorname{ind} n$ the index of n, the characters are $\chi_j(n) = e(j \operatorname{ind} n/(q-1))$ for $1 \le j \le q-2$, and χ_j is odd exactly when j is odd. Since $S(x) = \sum_{n \le x} \chi(n)$ is constant on each interval $[n, n+1)$ and periodic with $S(q) = 0$, the maximum of $|S(x)|$ is the maximum of $|S(n)|$ over $1 \le n \le q-1$, which the program computes in double precision; the accumulated rounding error is of order $q \cdot 10^{-16}$, far below the resolution of the statements made here. This is 1227 moduli and 5 733 937 characters. The programs, written in C, are archived with the paper (see the declarations), and a second program, written independently in Julia, recomputes the largest $M(\chi)/\sqrt{q}$ for every one of the 1227 moduli and agrees to six decimals.

Write

$$r(\chi) = \frac{M(\chi)}{\sqrt{q} \log\log q}.$$

Table 1 summarises the data by ranges of q, Figure 4 shows, for each q, the largest value of $r(\chi)$ over odd and over even characters, and Figure 5 is the distribution of $M(\chi)/\sqrt{q}$ for the 1 636 638 characters with $8419 \le q \le 9973$.

range of q	moduli	characters	odd max	even max	mean $M/\sqrt{q}$	$r > \frac{1}{2}$	$r > \frac{3}{4}$
5–1000	166	75 790	1.329	0.940	0.695	10 176	326
1001–2000	135	200 653	0.838	0.461	0.701	18 986	201
2001–3000	127	316 519	0.835	0.469	0.703	25 260	203
3001–4000	120	419 444	0.842	0.436	0.704	30 252	190
4001–5000	119	534 391	0.828	0.464	0.705	35 798	172
5001–6000	114	626 370	0.828	0.454	0.705	39 142	218
6001–7000	117	760 503	0.830	0.440	0.706	45 298	212
7001–8000	107	802 881	0.789	0.451	0.706	45 607	195
8001–9000	110	935 004	0.833	0.455	0.706	51 348	185
9001–10000	112	1 062 382	0.814	0.451	0.706	56 327	229
total	1227	5 733 937	1.329	0.940		358 194	2 131

Table 1. Prime moduli $5 \le q < 10^4$. The columns "odd max" and "even max" give the largest $r(\chi)$ over odd and over even characters in the range; the last two columns count characters.

The data show four things.

The maxima. For $q > 1000$ the largest odd value of $r(\chi)$ lies in $[0.714, 0.842]$ and the largest even value in $[0.378, 0.469]$. Only six characters in the whole range have $r(\chi) > 1$, two of them have $r(\chi) > 1.25$, none has $r(\chi) > 1.5$, and all six belong to moduli below 12, where $\log\log q < 1$. The largest value of $M(\chi)/\sqrt{q}$ is 1.839, at $q = 8951$; the largest value over even characters is 1.003.

Parity. For each of the 1227 moduli the largest $M(\chi)$ is attained by an odd character. Granville and Soundararajan proved, assuming the generalised Riemann hypothesis, that $M(\chi) \le (2C_\chi + o(1))\sqrt{q} \log\log q$ with

$$C_\chi = \frac{e^\gamma}{\pi} = 0.5669\ldots \ (\chi \text{ odd}), \qquad C_\chi = \frac{e^\gamma}{\sqrt{3}\,\pi} = 0.3273\ldots \ (\chi \text{ even}),$$

and conjectured that C_χ is the true constant [8, Theorem 6 and Conjecture 1].[3] Both maxima in Figure 4 lie between C_χ and $2C_\chi$, and the mean over $q > 1000$ of the ratio of the even maximum to the odd maximum is 0.546, a little below $1/\sqrt{3} = 0.577$. At $q \approx 10^4$ one has

[3]For even χ one has $S(q-1-x) = -S(x)$, so $S(x)$ is minus one half of the sum over $x < n < q-1-x$; the bound for even characters is the bound of [8] for sums over intervals, divided by two.

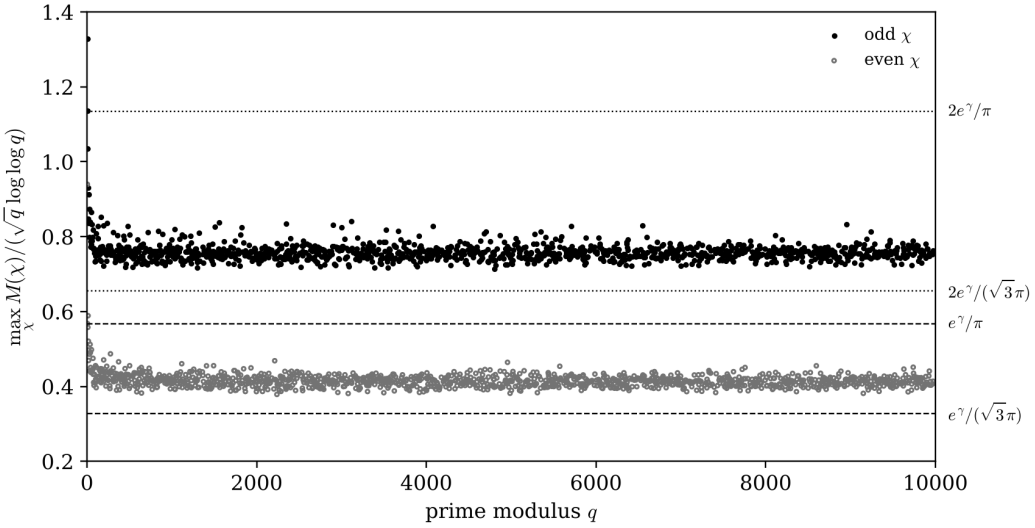


FIGURE 4. For each prime $q < 10^4$, the largest $r(\chi)$ over odd characters (upper cloud) and over even characters (lower cloud), with the constants $C_{\text{odd}} = e^\gamma/\pi$, $2C_{\text{odd}}$, $C_{\text{even}} = e^\gamma/(\sqrt{3}\pi)$ and $2C_{\text{even}}$.

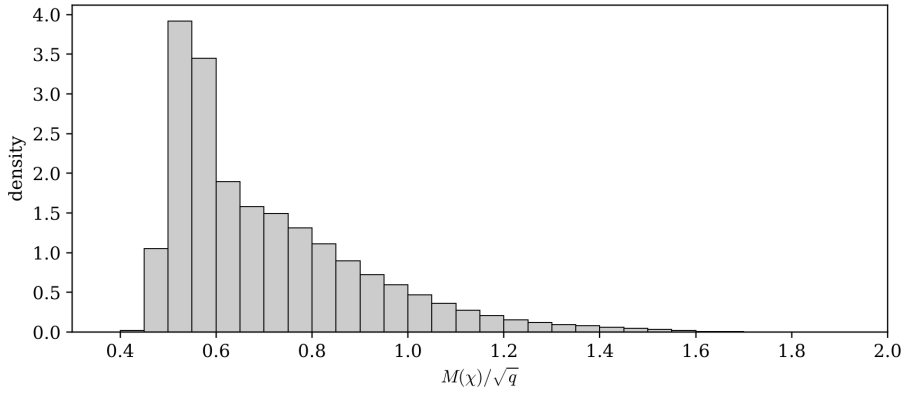


FIGURE 5. Distribution of $M(\chi)/\sqrt{q}$ over all nonprincipal characters to prime moduli $8419 \le q \le 9973$, in bins of width 0.05.

$\log\log q \approx 2.22$, so terms of relative size $1/\log\log q$ are still large, and the data neither confirm nor contradict the value C_χ.

The bulk. The mean of $M(\chi)/\sqrt{q}$ is 0.70 in every range and grows only in the third decimal, and every character in the range has $M(\chi) \ge 0.4\sqrt{q}$. The distribution in Figure 5 has a sharp left edge, a mode in $[0.50, 0.55)$, and a tail that decreases by a roughly constant factor per bin up to 1.5 and much faster beyond it, which is consistent with the doubly exponential decay found in [5].

The tail. The proportion of characters with $r(\chi) > \frac{1}{2}$ falls from 13.4% for $q \le 1000$ to 5.3% for $9000 < q < 10^4$, and the proportion with $r(\chi) > \frac{3}{4}$ falls from 0.43% to 0.022%. Corollary 1.2 bounds the number of characters above $\tau\sqrt{q}\log\log q$ by $q^{c_1/\tau}$ up to a power of $\log q$; for $9000 < q < 10^4$ the data give about 500 characters per modulus with $r(\chi) > \frac{1}{2}$ and about two with $r(\chi) > \frac{3}{4}$, so the true tail is far thinner than the bound, as one expects from a zero-density argument.

8. Open problems

Table 2 lists the statements about $M(\chi)$ and $L'/L(1,\chi)$ that bear on this paper, with their status. "Proved" means proved unconditionally; "GRH" means proved under the generalised Riemann hypothesis; "open" means that no proof is known, with or without the present results.

Statement	Status	What this paper adds
$M(\chi) \ll \sqrt{q}\log q$ for every χ	proved (1918)	nothing
$M(\chi) \gg \sqrt{q}\log\log q$ for infinitely many quadratic χ	proved [13]	nothing
$M(\chi) \ll \sqrt{q}\log\log q$ for every χ	GRH [12]; open	proved for all but $O((\log q)^{c_3}q^{\varepsilon})$ characters, constant c_1/ε (Theorem 1.1)
$M(\chi) = o(\sqrt{q}\log q)$ for every χ	open	proved for all but $\exp(O((\log\log q)^2))$ characters (Corollary 1.2)
$M(\chi) \le (2C_\chi + o(1))\sqrt{q}\log\log q$	GRH [8]	numerical maxima below $2C_\chi$ for $10^3 < q < 10^4$ (Section 7)
$M(\chi) \le (C_\chi + o(1))\sqrt{q}\log\log q$	open [8]	data between C_χ and $2C_\chi$; inconclusive
Frequency of $M(\chi) > \tau\sqrt{q}$ for $\tau \le \log\log q - C$	proved [5]	tail bound for all larger thresholds (Corollary 1.2)
$M(\chi) \ll \sqrt{q}(\log q)^{1-\delta}$ for χ of fixed odd order	proved [8, 7, 10]	nothing
$\lvert L'/L(1,\chi)\rvert \le (2+o(1))\log\log q$	GRH [9]	$\ll \varepsilon^{-1}\log\log q$ for all but $O_\varepsilon(q^{\varepsilon+o(1)})$ characters (Theorem 1.5)
Zero-free half-plane $\operatorname{Re} s > \theta$, $\theta < 1$, for all $L(s,\chi)$	open	would give both bounds for every χ with constant $\asymp (1-\theta)^{-1}$ (Corollary 1.4)

Table 2. Statements about large character sums and their status.

Three questions are suggested by the method.

Fewer exceptions. The exceptional set of Theorem 1.1 is controlled by the number of characters χ such that some twist $\chi\psi$ with a modulus of ψ at most $(\log q)^{16}$ has a zero with real part above $1-\varepsilon/8$ and height at most q. In the zero-density estimate (2.1) the twists have modulus $k = qr$ and height $T = q$, so the quantity kT is about q^2; a smaller exponent in place of $2(1-\sigma)/\sigma$, or a version of Proposition 4.1 that needs the zero-free region only up to height $(\log q)^A$, would lower the exponent of q in Theorem 1.1 in proportion. The second change alone would halve it. In the present proof the height q comes from the truncation $T = x$ of the explicit formula in Lemma 3.1, applied to prime sums of length up to q.

The constant. The proof gives c_1 in terms of the constants of the zero-density estimate (2.1) and the Montgomery–Vaughan estimate (2.2), neither of which is explicit in the sources. With explicit versions of both, Theorem 1.1 becomes a statement with numbers, which could be set against Table 1.

Even characters. For every modulus in our range the maximum is odd, and the ratio of the even maximum to the odd maximum is close to $1/\sqrt{3}$. It would be interesting to prove, for a positive proportion of prime moduli and without hypothesis, that the largest $M(\chi)$ is attained by an odd character.

The large sieve and zero-free regions are used for related questions in [3, 4], and approximations connected with the Riemann hypothesis and with Ramanujan's τ-function are studied in [2, 1].

Appendix A. Small moduli

Table 3 lists, for each prime $5 \le q < 200$, the character with the largest $M(\chi)$. The characters are numbered as in Section 7: g is the least primitive root, $\chi_j(g) = e(j/(q-1))$, the order of χ_j is $(q-1)/\gcd(j, q-1)$, and x is the least integer with $|S(x)| = M(\chi_j)$. Since $\chi_{q-1-j} = \bar{\chi}_j$ has the same partial sums up to conjugation, only the smaller index of each conjugate pair is listed.

q	g	j	order	x	x/q	$M/\sqrt{q}$	$r(\chi)$	q	g	j	order	x	x/q	$M/\sqrt{q}$	$r(\chi)$
5	2	1	4	2	0.400	0.6325	1.3290	97	5	15	32	48	0.495	1.1149	0.7332
7	3	1	6	3	0.429	0.7559	1.1355	101	2	45	20	50	0.495	1.1232	0.7344
11	2	5	2	5	0.455	0.9045	1.0342	103	5	7	102	34	0.330	1.1482	0.7487
13	2	3	4	6	0.462	0.8771	0.9311	107	2	3	106	42	0.393	1.1780	0.7641
17	3	3	16	8	0.471	0.8821	0.8470	109	6	27	4	54	0.495	1.2489	0.8079
19	2	7	18	9	0.474	0.9037	0.8368	113	3	9	112	45	0.398	1.1654	0.7502
23	5	11	2	9	0.391	1.0426	0.9123	127	3	7	18	54	0.425	1.2540	0.7948
29	2	5	28	14	0.483	0.9692	0.7983	131	2	65	2	65	0.496	1.3106	0.8273
31	3	15	2	10	0.323	1.0776	0.8735	137	3	13	136	63	0.460	1.2160	0.7632
37	2	11	36	18	0.486	1.0770	0.8388	139	2	3	46	64	0.460	1.2219	0.7655
41	6	3	40	16	0.390	1.0330	0.7874	149	2	5	148	59	0.396	1.2119	0.7526
43	3	3	14	18	0.419	1.0241	0.7730	151	6	13	150	58	0.384	1.1905	0.7381
47	5	23	2	18	0.383	1.1669	0.8656	157	5	21	52	72	0.459	1.2514	0.7722
53	2	3	52	21	0.396	1.0744	0.7792	163	2	3	54	69	0.423	1.1974	0.7355
59	2	29	2	29	0.492	1.1717	0.8337	167	5	83	2	66	0.395	1.3929	0.8531
61	2	1	60	24	0.393	1.1161	0.7895	173	2	9	172	57	0.329	1.2423	0.7576
67	2	17	66	33	0.493	1.1477	0.7991	179	2	5	178	81	0.453	1.2639	0.7677
71	7	35	2	20	0.282	1.1868	0.8185	181	2	3	60	72	0.398	1.2766	0.7744
73	5	9	8	24	0.329	1.1178	0.7675	191	19	95	2	81	0.424	1.3748	0.8289
79	3	1	78	33	0.418	1.1340	0.7690	193	5	7	192	96	0.497	1.1998	0.7225
83	2	3	82	27	0.325	1.1361	0.7646	197	2	1	196	78	0.396	1.2605	0.7572
89	3	1	88	44	0.494	1.1356	0.7563	199	3	99	2	66	0.332	1.2760	0.7657

Table 3. The character with the largest $M(\chi)$ for each prime $5 \le q < 200$; every one is odd.

Three features are visible already at this height. The maximising character is odd in every case. For $q \equiv 3 \pmod 4$ it is often the quadratic character $\chi_{(q-1)/2}$, of order 2 ($q = 23, 47, 59, 71, 131, 167, 191, 199$); for $q > 1000$ this happens for 59 of the 532 primes $q \equiv 3 \pmod 4$, and for 1001 of the 1061 primes $q > 1000$ the maximising character has order larger than 12. The point x/q where the maximum is reached lies between 0.28 and 0.50 for every $q < 200$; by the symmetry $|S(q-1-x)| = |S(x)|$ of odd characters, the half $x \le q/2$ suffices.

Appendix B. The largest values for $9000 < q < 10^4$

Table 4 gives, for each of the 112 primes $9000 < q < 10^4$, the largest $r(\chi)$ over odd and over even characters, the mean of $M(\chi)/\sqrt{q}$ over all $q-2$ nonprincipal characters, and the numbers of characters with $r(\chi) > \frac{1}{2}$ and $r(\chi) > \frac{3}{4}$.

Appendix C. The constants

All constants are absolute. The table records where each one enters; none of them is computed explicitly, and the proofs use only that they exist.

The only place where the width $1-\theta$ of the zero-free rectangle enters the transfer theorem is the factor $(1-\theta)^{-1}$ in Lemma 3.3, which comes from the sum of $m^{-\theta}$ over the cofactors $m < t/Y$, and the length $A = \lceil 40/(1-\theta) \rceil$ of the smooth range, which costs

q	odd	even	mean	$>\frac{1}{2}$	$>\frac{3}{4}$	q	odd	even	mean	$>\frac{1}{2}$	$>\frac{3}{4}$
9001	0.749	0.421	0.706	472	0	9473	0.753	0.399	0.706	492	4
9007	0.751	0.424	0.706	496	2	9479	0.753	0.413	0.706	477	2
9011	0.772	0.409	0.706	482	2	9491	0.737	0.401	0.706	495	0
9013	0.770	0.418	0.706	492	2	9497	0.759	0.396	0.707	516	2
9029	0.755	0.426	0.706	494	2	9511	0.725	0.415	0.706	511	0
9041	0.757	0.417	0.706	496	2	9521	0.763	0.431	0.706	490	4
9043	0.766	0.404	0.706	488	6	9533	0.755	0.412	0.706	516	2
9049	0.730	0.424	0.706	496	0	9539	0.764	0.400	0.706	491	3
9059	0.774	0.411	0.706	451	2	9547	0.745	0.403	0.706	498	0
9067	0.784	0.399	0.706	494	4	9551	0.771	0.422	0.706	523	5
9091	0.755	0.406	0.706	490	2	9587	0.744	0.410	0.706	522	0
9103	0.780	0.410	0.706	487	4	9601	0.727	0.407	0.706	510	0
9109	0.755	0.422	0.707	494	4	9613	0.782	0.410	0.706	494	2
9127	0.726	0.416	0.706	495	0	9619	0.772	0.422	0.706	514	2
9133	0.741	0.426	0.706	480	0	9623	0.753	0.401	0.706	485	2
9137	0.748	0.407	0.706	494	0	9629	0.762	0.446	0.706	504	2
9151	0.770	0.382	0.706	479	6	9631	0.754	0.412	0.706	481	4
9157	0.746	0.418	0.706	480	0	9643	0.744	0.409	0.706	506	0
9161	0.778	0.414	0.706	498	2	9649	0.745	0.431	0.706	506	0
9173	0.746	0.414	0.706	496	0	9661	0.747	0.414	0.706	512	0
9181	0.753	0.425	0.706	504	2	9677	0.766	0.410	0.706	494	2
9187	0.794	0.410	0.706	502	2	9679	0.777	0.385	0.707	519	4
9199	0.761	0.398	0.706	486	4	9689	0.754	0.423	0.706	514	2
9203	0.747	0.422	0.706	494	0	9697	0.772	0.416	0.707	524	2
9209	0.764	0.391	0.706	510	6	9719	0.778	0.390	0.707	515	5
9221	0.745	0.434	0.706	488	0	9721	0.724	0.407	0.707	510	0
9227	0.764	0.403	0.706	506	2	9733	0.751	0.415	0.706	506	2
9239	0.814	0.398	0.706	487	5	9739	0.790	0.451	0.706	516	2
9241	0.741	0.414	0.706	494	0	9743	0.770	0.435	0.706	527	3
9257	0.769	0.401	0.707	496	4	9749	0.755	0.427	0.706	520	2
9277	0.746	0.423	0.706	500	0	9767	0.766	0.410	0.706	485	4
9281	0.754	0.428	0.706	494	2	9769	0.764	0.425	0.706	516	4
9283	0.780	0.395	0.706	478	2	9781	0.777	0.397	0.706	490	4
9293	0.760	0.406	0.706	512	4	9787	0.771	0.423	0.706	484	2
9311	0.745	0.417	0.706	493	0	9791	0.775	0.409	0.706	513	6
9319	0.752	0.411	0.706	506	2	9803	0.735	0.423	0.707	523	0
9323	0.746	0.416	0.706	514	0	9811	0.744	0.423	0.706	512	0
9337	0.760	0.387	0.706	472	4	9817	0.762	0.410	0.706	526	2
9341	0.751	0.412	0.706	498	2	9829	0.758	0.413	0.707	526	2
9343	0.757	0.400	0.706	496	4	9833	0.730	0.423	0.706	474	0
9349	0.757	0.420	0.706	488	2	9839	0.765	0.406	0.706	523	2
9371	0.759	0.415	0.706	495	2	9851	0.760	0.395	0.706	517	2
9377	0.753	0.419	0.706	516	2	9857	0.754	0.414	0.706	510	2
9391	0.748	0.401	0.706	505	0	9859	0.758	0.398	0.706	518	2
9397	0.753	0.427	0.706	508	2	9871	0.749	0.401	0.706	528	0
9403	0.762	0.413	0.706	512	2	9883	0.780	0.406	0.707	506	2
9413	0.760	0.419	0.706	514	4	9887	0.744	0.426	0.706	529	0
9419	0.754	0.416	0.706	484	4	9901	0.768	0.411	0.707	512	2
9421	0.755	0.414	0.706	498	2	9907	0.767	0.406	0.707	512	4
9431	0.745	0.411	0.706	491	0	9923	0.749	0.428	0.707	528	0
9433	0.751	0.407	0.706	488	2	9929	0.756	0.441	0.706	542	2
9437	0.776	0.425	0.706	502	2	9931	0.749	0.404	0.706	542	0
9439	0.774	0.408	0.706	503	2	9941	0.755	0.413	0.706	532	2
9461	0.790	0.405	0.706	494	2	9949	0.756	0.403	0.706	534	2
9463	0.784	0.411	0.707	504	2	9967	0.753	0.407	0.706	542	2
9467	0.761	0.426	0.706	511	2	9973	0.762	0.414	0.706	518	6

TABLE 4. Primes $9000 < q < 10^4$: largest $r(\chi)$ over odd and over even characters, mean of $M(\chi)/\sqrt{q}$, and the numbers of characters with $r(\chi)$ above $\frac{1}{2}$ and $\frac{3}{4}$.

constant	origin
c_8	number of zeros of $L(s,\xi)$ with $\lvert\gamma - t\rvert \le 1$, from the partial fraction formula (Z1)
c_9	zero-density estimate (2.1), $\sigma \ge 4/5$
c_{10}	exponential sums with multiplicative coefficients (2.2)
c_{11}	sum of $1/n$ over Y-smooth n, Lemma 2.3
c_{12}	Stirling's formula, Lemma 2.4
c_{13}	prime sums from a zero-free rectangle, Lemmas 3.1–3.3; independent of θ
$c_{14} = 164c_{11} + 3c_{10} + 2$	Proposition 4.1
$c_4 = \max(c_{14} + 2, (3 + L_0)/\log\log 3)$	Theorem 1.3
$c_1 = 8c_4$, $c_2 = 2^{37}3^{14}c_9$, $c_3 = 51$	Theorem 1.1
c_{15}, c_{16}, T_0	logarithmic derivative, Proposition 6.1
$c_5 = 20c_{16}$, c_6, c_7	Theorem 1.5
L_0	threshold for $\log q$ beyond which the numerical inequalities of Section 4 hold

TABLE 5. The constants and where they arise.

$\log Y \le \frac{82}{1-\theta}\log\log q$. Both are linear in $(1-\theta)^{-1}$, and this is why Theorem 1.1 holds with the constant c_1/ε and not with a higher power of $1/\varepsilon$.

APPENDIX D. THE COMPUTATIONS

Character sums. Two independent programs compute $M(\chi)$: a C program, which produced every number in Section 7 and Appendices A–B, and a Julia program written separately, which recomputes the largest $M(\chi)/\sqrt{q}$ for every prime $5 \le q < 10^4$ and agrees with the C output to six decimals for all 1227 moduli. Both use the least primitive root, the table of indices and complex double precision; the partial sums are accumulated in order, so that the error after q steps is of order q times the machine precision.

Finite arithmetic. A Lean 4 file, using the core library only, checks the integer facts used in the proofs: the exponent count $16 \cdot 2/7 < 5$ and $32 + 5 + 14 = 51$ of Section 5, the value $2^{37}3^{14} = 657\,366\,253\,849\,018\,368$, the reduction of $2(1-\theta)/\theta \le \frac{16}{7}(1-\theta)$ to $\theta \ge 7/8$, the bound $3 + 2\pi R \le 8R$ for $R \ge 2$ with $\pi < 22/7$, the exponents $16 + 16 + 2 + 6 = 40$ and constants $8 \cdot 9 = 72$, $2 \cdot 82 = 164$ of Section 4, $5 \cdot 4 = 20$ of Section 6, and the totals of Table 1. Every theorem is checked by the kernel; the file uses no axiom beyond the two standard ones of Lean's core, propositional extensionality and quotient soundness, and some of its theorems use none.

The Gamma bound. The Julia program also evaluates $\lvert\Gamma(u + it)\rvert(1 + t^2)/15$ on a grid of step 0.005 in $u \in [3/4, 2]$ and 0.01 in $t \in [0, 40]$; the largest value is 0.116, well below the value 1 allowed by Lemma 2.4, whose proof is in any case analytic.

AUTHOR CONTRIBUTIONS

Deep Bhattacharjee is the sole author. He posed the problems in the form treated here, proved the results, wrote the paper, drew the figures, ran the computations of Section 7, and wrote the Lean 4 file and the Julia programs that check the numerical statements.

OPEN GLUON AND ANAXIA

Two tools built by the author organised this project. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each

construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped. It was used here to keep the chain of constants $c_8, \dots, c_{16}$ and the data of Section 7 with their checks in one record. Every statement of this paper is proved in the text.

Declarations

Data and code availability. The C programs that compute $M(\chi)$ for all characters to prime moduli below 10^4 and their output, the Julia programs that recompute the largest values for every modulus, the data of Appendix A and the Gamma bound, the script that draws the figures, and the Lean 4 file of Appendix D are in the public repository `https://github.com/creelie/Breakthrough-problems-3`.

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